

1. Administrative & Study Identification

Full Study Title: A Phase II, Multicenter, Double-Blind, Placebo-Controlled Study to Assess the Efficacy and Safety of R07790121 in Participants With Moderate to Severe Rheumatoid Arthritis Who Have an Inadequate Response or Intolerance to TNF and/or JAK Inhibitors **Short Title/Acronym:** dRAvite Study / Afimkibart for Rheumatoid Arthritis **Protocol Number:** WA45846 / 2025-521034-28-00 **NCT Number:** NCT07137598 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Afimkibart (R07790121, RVT-3101, PF-06480605) **Phase of Development:** Phase II **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead (global.roche.genentech.trials@roche.com)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of Afimkibart (R07790121) compared with a placebo in participants with moderate to severe rheumatoid arthritis (RA) who have an inadequate response or intolerance to TNF and/or JAK inhibitors. **Secondary Objectives:** To evaluate disease activity (e.g., changes in the DAS28 CRP score), long-term safety, incidence of adverse events (including injection site reactions), and overall clinical remission rates.

2.2 Background & Rationale Rheumatoid arthritis (RA) is an autoimmune and inflammatory condition causing painful swelling in the joints. While several treatments exist—such as TNF and JAK inhibitors—many patients do not achieve remission, lose response over time, or cannot tolerate them. Afimkibart is a novel anti-TL1A monoclonal antibody that offers a new approach by targeting specific immune pathways involved in the inflammation process. This study is necessary to determine if targeting TL1A can effectively and safely control the disease in patients who have exhausted traditional biologic or targeted synthetic therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1 Randomized (Participants in the placebo group will cross over to receive Afimkibart halfway through the study)

3.2 Planned Enrollment & Duration Target **NS:** 160 participants **Number of Sites:** 23 locations (Including the US, Denmark, France,

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Diagnosis of RA for ≥ 3 months, fulfilling the 2010 ACR/EULAR classification criteria for RA. 3. Moderate to severe active RA, defined by the presence of ≥ 6 swollen joints and ≥ 6 tender joints at screening and baseline (based on 66/68-joint count). 4. Demonstrated an inadequate response, loss of response, or intolerance to ≥ 1 conventional synthetic disease-modifying antirheumatic drug (csDMARD) and to TNF and/or JAK inhibitors.

4.2 Exclusion Criteria 1. Have failed more than two TNF inhibitors or JAK inhibitors; or Class IV RA according to ACR revised response criteria. 2. History of active Hepatitis B (HBV), Hepatitis C (HCV), or HIV infection, or severe chronic/recurrent infections. 3. Past or current use of other biologic disease-modifying antirheumatic drugs (bDMARDs) (excluding TNF inhibitors) or rituximab, or treatment with intra-articular/systemic corticosteroids in the preceding 8 weeks.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered via subcutaneous (SC) injection. Injections are administered more frequently at the beginning of the study, with intervals decreasing during the maintenance phase. **Route of Administration:** Subcutaneous (SC) **Duration of Treatment:** Total study duration through to October 2027 (approximately 96 weeks of observation/treatment).

5.2 Concomitant & Prohibited Therapy Permitted: Stable background treatment with eligible csDMARDs. **Prohibited:** Concurrent investigational therapies, rituximab, or non-TNF bDMARDs.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Joint Count, Lab screening for viral markers
Baseline	Day 0	Randomization, First SC Injection, Baseline DAS28 CRP assessment
Interim	Scheduled Cadence	SC Injections, Efficacy Assessment (DAS28), Safety monitoring (Injection site checks)
End of Study	Study Completion	Final Efficacy Assessment, Exit Interview, Safety follow-up

(Note: Patients initially receiving placebo will cross over to receive active drug midway through the study).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants achieving clinical response and evaluating the efficacy and safety of Afimkibart against placebo. **Measurement Method:** Clinical joint evaluations (e.g., 66/68 joint counts), patient-reported outcomes, and central laboratory analysis.

7.2 Secondary & Exploratory Endpoints 1. Changes from baseline in the DAS28 CRP score (incorporating swollen/painful joints, physician/patient global assessments, and C-Reactive Protein levels). 2. Incidence and severity of Treatment-Emergent Adverse Events (TEAEs) and Serious Adverse Events (SAEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of injection site reactions (bruising, swelling, pain) and signs of infection or systemic reactions. **Serious Adverse Events (SAEs):** Standard reporting timelines (24 hours) for severe events such as anaphylaxis or severe infections. **Data Monitoring Committee (DMC):** External/Internal committee details mandated by the Sponsor, Hoffmann-La Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy measures; Safety Set for all participants who receive at least one dose of the investigational product. **Sample Size Justification:** N=160, sufficiently powered to detect a statistically significant difference in clinical response parameters (such as DAS28-CRP or ACR responses) versus placebo. **Handling of Missing Data:** Standard imputation (e.g., Non-Responder Imputation for binary clinical response endpoints).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard risk-based onsite and remote monitoring of participating clinical sites across the US and Europe. **Audit Trail:** Validated eCRF tracking and query resolution via standard data clarification forms.

11. Study Identifiers & Governance

Primary ID: WA45846 **Registry Number:** NCT07137598 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** dRAvite Study / Afimkibart for Rheumatoid Arthritis **Official Regulatory Title:** A Phase II, Multicenter, Double-Blind, Placebo-Controlled Study to Assess the Efficacy and Safety of R07790121 in Participants With Moderate to Severe Rheumatoid Arthritis Who Have an Inadequate Response or Intolerance to TNF and/or JAK Inhibitors

12. Clinical Framework (Rheumatoid Arthritis Specific)

Primary Condition: Moderate to Severe Rheumatoid Arthritis (RA) **Diagnosis Threshold:** ≥ 6 swollen joints and ≥ 6 tender joints **Medical Methodology:** Targeted biologic therapy (TL1A Inhibition) **Optimization Target:** Disease remission and reduction of DAS28 CRP inflammation markers

13. Study Procedures & Methodology

Management Pathway: Step-up therapy for Treatment-Resistant RA **Education Protocol (HCP):** Subcutaneous (SC) injection protocols and blinding maintenance **Education Protocol (Patient):** Injection site reaction awareness, logging of daily symptom severity **Quality Audit Mechanism:** Independent central reading of laboratory markers (CRP) and standard clinical oversight

14. Observation & Follow-Up Schedule

Total Duration: Follow-up estimated until October 2027 (approx. 96 weeks overall) **Onsite Visit Cadence:** Per protocol administration schedule for SC injections **Remote Monitoring Frequency:** Standard protocol safety calls **Checklist Review Interval:** Regular review of patient-reported joint pain and swelling

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				